MASTER IN COMPUTER SCIENCE

Distributed Deep Learning Systems

Project Proposal

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1 Paper choice : Adaptive LoRA

We've chosen the Adaptive LoRA Experts Allocation and Selection for Federated Fine-Tuning [1] paper for our project.

The fine-tuning of **L**arge **L**anguage **M**odels (LLMs) in **F**ederated **L**earning (FL) is is difficult because client data can come from very different domains, such as medicine or finance. Although **Lo**w-**R**ank **A**daptation (LoRA) Lowers computation by updating only small low rank matrices, one shared LoRA module cannot properly learn the different knowledge needed for distinct tasks.
Innovation 1: Adaptive LoRA Expert Allocation
FedLEASE addresses this problem by grouping similar clients so they can jointly train domain-specific experts. First, clients do a short local training phase. Then, the server compares the **B** matrices, which contain more task-specific information than the **A** matrices, using **Agglomerative Hierarchical Clustering**. With the **silhouette coefficient**, FedLEASE automatically finds the best number of experts (**M**), while balancing specialization and knowledge sharing.
Innovation 2: Adaptive top-M Selection Mechanism
To manage diversity at the sample level, FedLEASE introduces a dynamic routing method instead of the usual fixed **top-k** selection. The router output space is extended to $R^{(2M-1)}$, which ensures that the client's own expert is always selected for stable training. At the same time, it can also use other experts depending on the input data, without needing manual hyperparameter tuning.

2 What we plan to do

We know from the paper's limitations in section E, that one issue with this model is it's applicability with dynamic clusters. In our project we'll attempt to make an implementation of Adaptive LoRA compatible with dynamic clustering.

In it's current state the Adaptive LoRA Experts Allocation and Selection for FFT assumes a a static client population and a fixed expert assignments throughout the training process. Such models don't truly represent what appends in real situations where both client availability and data distribution evolve over time.
The main improvements dynamic clustering could bring in a non-stationary setting are : **(1) A better adaptation overtime**, with client data distribution evolving you want to be sure that clients are matched with the right expert and not an outdated one. **(2) Reduce negative transfers**, by preventing un-related clients from sharing experts for the co-training. **(3) Expert budget optimization**, the FedLEASE model already makes a choice for expert allocation but it would need to be reworked for a non-stationary setting. **(4) potential robustness and fairness gains**, rare values or evolving one are more likley to get stuck in an assignment. By allowing reassignment these situations can be fixed.

2.1 Our addition to the model

We will first reproduce the original FedLEASE setup. We will then extend it with a dynamic reclustering mechanism during training, instead of fixing client-to-expert assignments only once at initialization. Based on the LoRA B matrices already used in the paper for similarity, we can update the clusters over time to better adapt to changing client participation and data distributions. And then compare the original static version with our dynamic version in non-stationary federated settings. Once we have compared them, we will be able to measure whether this improves performance without costing too much.

References

[1] Lei Wang; Jieming Bian; Letiang Zhang; Jie Xu. Adaptive lora experts allocation and selection for federated fine-tuning, 2025. Original project paper. URL: https://arxiv.org/pdf/2509.15087.